Supplemental Material for "Seed geometry in nucleation-controlled transition times in time-dependent Ginzburg–Landau models"

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SUPPLEMENTARY FIGURES

ORDERED-LIKE SEED STATISTICS

Figure S1 shows the statistics of ordered-like seeds, defined by

$$|\phi(\mathbf{x}, 0)| > 0.5\phi_s. \tag{1}$$

These quantities remain close to zero for all initial-state labels, indicating that the initial state does not typically contain pre-existing stable-phase droplets.

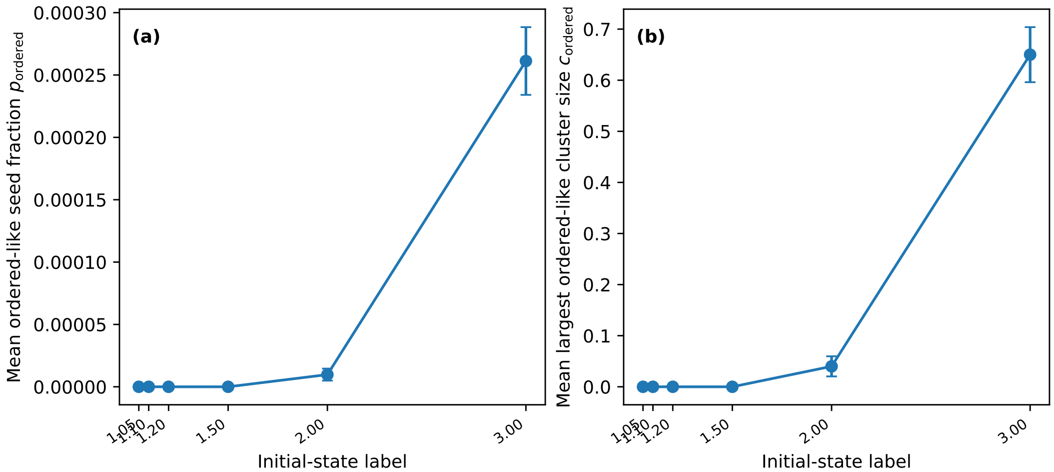

FIG. S1. Ordered-like seed statistics in the initial state. (a) Mean ordered-like seed fraction, defined by $|\phi(\mathbf{x}, 0)| > 0.5\phi_s$. (b) Mean largest ordered-like seed cluster size. Error bars indicate standard errors over stochastic realizations.

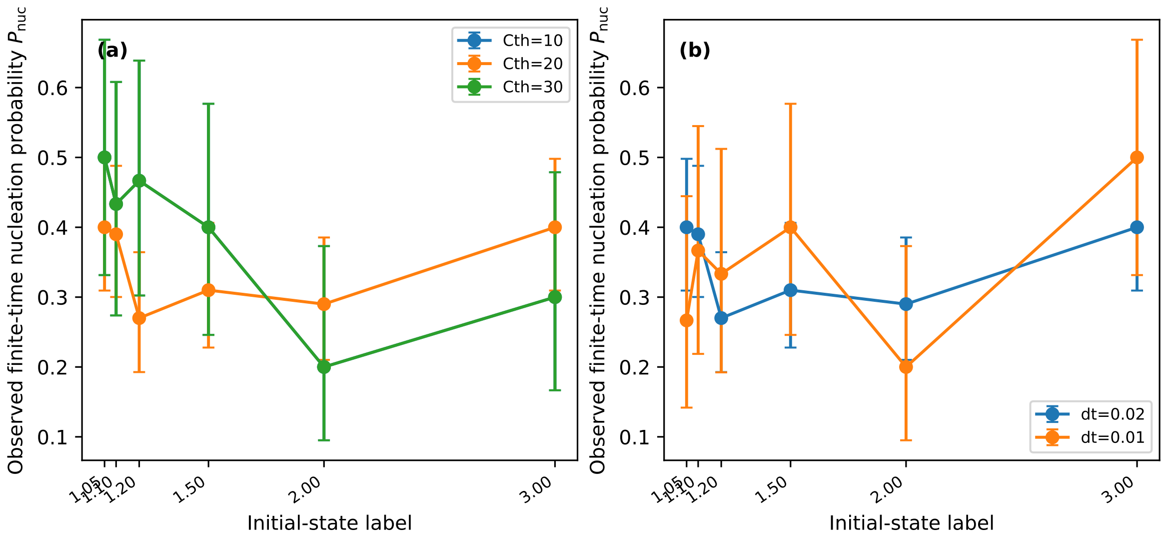

FIG. S2. Numerical robustness checks for the observed finite-time nucleation probability. (a) Dependence on the cluster threshold C_{th} used to identify nucleation events. (b) Dependence on the time step. Error bars indicate 95% Wilson binomial confidence intervals.

NUMERICAL ROBUSTNESS CHECKS

Figures S2 and S3 show numerical and operational robustness checks. The purpose of these checks is not to establish full finite-size convergence, but to verify that the central seed-quality trend is not an artifact of one particular numerical setting or nucleation criterion.

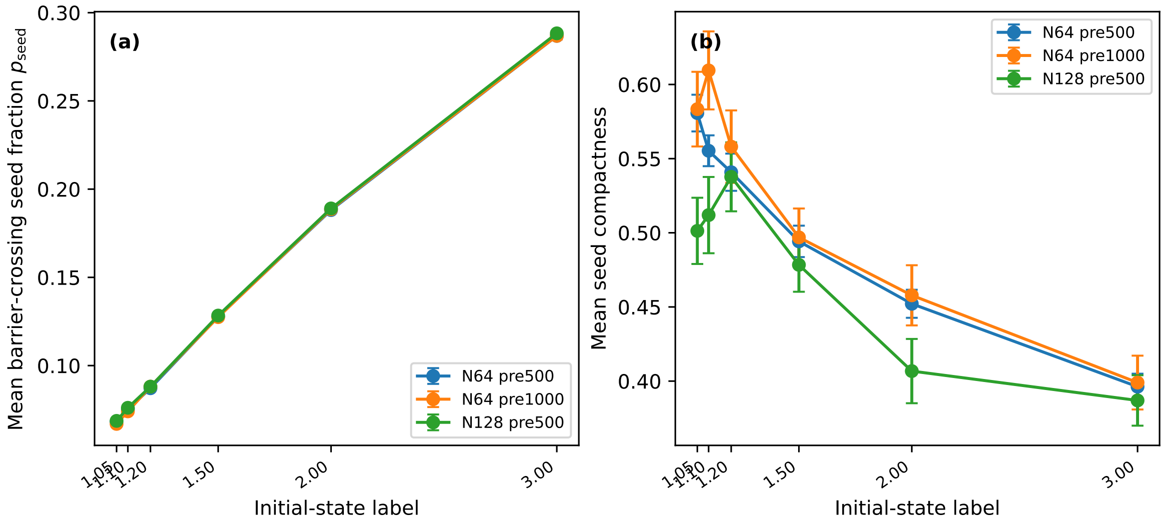

FIG. S3. Robustness of seed statistics against system size and pre-equilibration length. (a) Barrier-crossing seed fraction p_{seed}. (b) Seed compactness. The increase of p_{seed} and the decrease of seed compactness with the initial-state label are preserved in these checks. Error bars indicate standard errors over stochastic realizations.

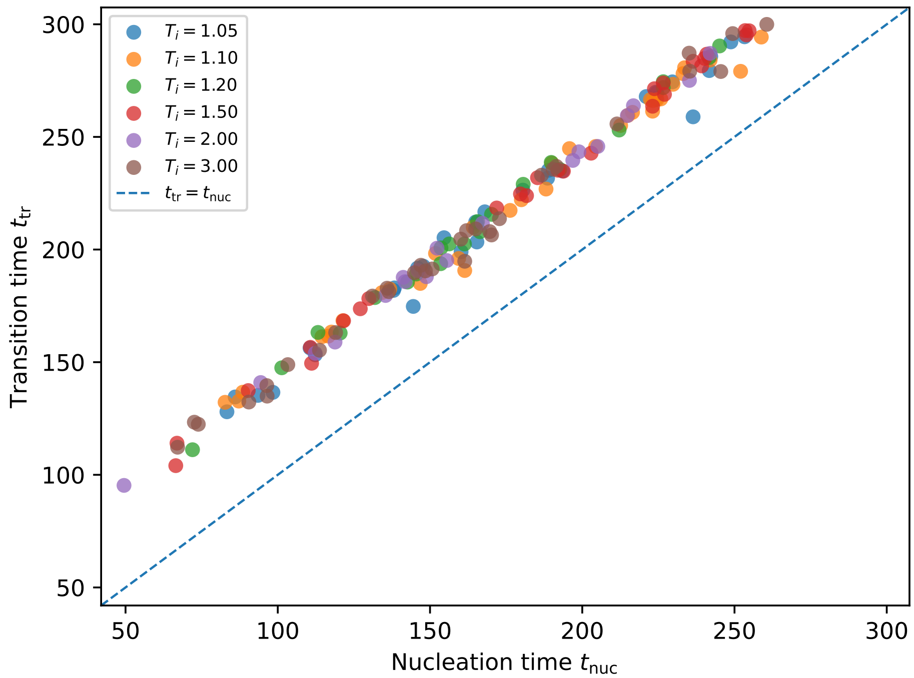

FIG. S4. Relation between the cluster nucleation time $t_{\rm nuc}$ and the global transition time $t_{\rm tr}$ in the first-order ϕ^6 model. Only samples for which both times were observed are shown. The dashed line indicates $t_{\rm tr} = t_{\rm nuc}$.

RELATION BETWEEN NUCLEATION AND TRANSITION TIMES

Figure S4 shows the relation between the cluster nucleation time $t_{\rm nuc}$ and the global transition time $t_{\rm tr}$ for samples in which both times were observed. The two times are strongly correlated. This supports the interpretation that the global transition time is closely tied to the stochastic nucleation time in the first-order model.

For the main first-order data set, the combined sample contains 600 realizations. Among these, 206 realizations show an observed cluster nucleation event and 165 realizations show an observed global transition by the final observation time. The scatter plot includes the 165 realizations for which both $t_{\rm nuc}$ and $t_{\rm tr}$ were observed. The Pearson correlation between these two times is approximately 0.997, and the Spearman rank correlation is approximately 0.995. The mean growth delay is $\langle \Delta t_{\rm grow} \rangle = \langle t_{\rm tr} - t_{\rm nuc} \rangle = 43.53 \pm 4.28$ time units, using the same summary convention as in the main text.

TABLE S1. Logistic-model comparison for the binary outcome of whether cluster nucleation was observed by the final observation time. Lower AIC is better. The sample contains 600 realizations, of which 206 nucleated by the final observation time.

Model	Predictors	ΔAIC
Seed quality	$C_{\rm comp}, R_g$	0.00
Intercept only	none	1.08
Label only	T_i	2.87
Amount plus quality	$p_{\rm seed}, c_{\rm seed}, C_{\rm comp}, R_g$	3.70
Seed amount	$p_{\rm seed}, c_{\rm seed}$	4.09
All predictors	$T_i, p_{\rm seed}, c_{\rm seed}, C_{\rm comp}, R_g$	5.47

SAMPLE-LEVEL SEED-GEOMETRY ANALYSIS

As an additional check, we examined whether initial seed variables carry sample-level information about whether a cluster nucleation event is observed within the finite observation window. For each realization in the main first-order data set, we defined a binary outcome equal to one if $t_{\rm nuc}$ was observed by the final observation time and zero otherwise. We then fitted logistic models using standardized predictors. This analysis should not be interpreted as a committor calculation: it uses only a small set of initial geometric descriptors and a finite final observation time.

Table S1 summarizes the Akaike information criterion (AIC) values for several low-dimensional models of the observed finite-time nucleation outcome. The model using seed compactness and radius of gyration has the lowest AIC among the tested models, but its improvement over the intercept-only model is modest, with $\Delta\mathrm{AIC} = 1.08$. The combined model containing both seed-amount and seed-geometry variables is not preferred by AIC. Thus, the sample-level predictive evidence is modest and should be used only as a consistency check.

Table S2 shows the standardized coefficients for the model containing both seed-amount and seed-geometry variables. Seed compactness has a positive standardized coefficient, $\beta = 0.25$, with an approximate two-sided p value of 0.052. The coefficients of $p_{\rm seed}$, $c_{\rm seed}$, and R_g are not statistically decisive in this finite data set. This result is consistent with the physical interpretation that compactness may be relevant, but it is not by itself a quantitative

TABLE S2. Standardized coefficients in the logistic model containing both seed-amount and seed-geometry variables. Approximate standard errors and p values are based on the local Hessian of the log-likelihood.

Predictor	Standardized coefficient	Approx. p value	Odds ratio per 1 s.d.
p_{seed}	-0.090	0.618	0.914
c_{seed}	0.122	0.662	1.130
C_{comp}	0.250	0.052	1.284
R_g	0.168	0.568	1.182

nucleation-rate theory and should not be read as a committor estimate.

Figure S5 visualizes the same conservative sample-level check. The binned probabilities and within-p_{seed} comparisons are intended only as descriptive diagnostics. They do not replace a committor analysis or a direct critical-droplet calculation.

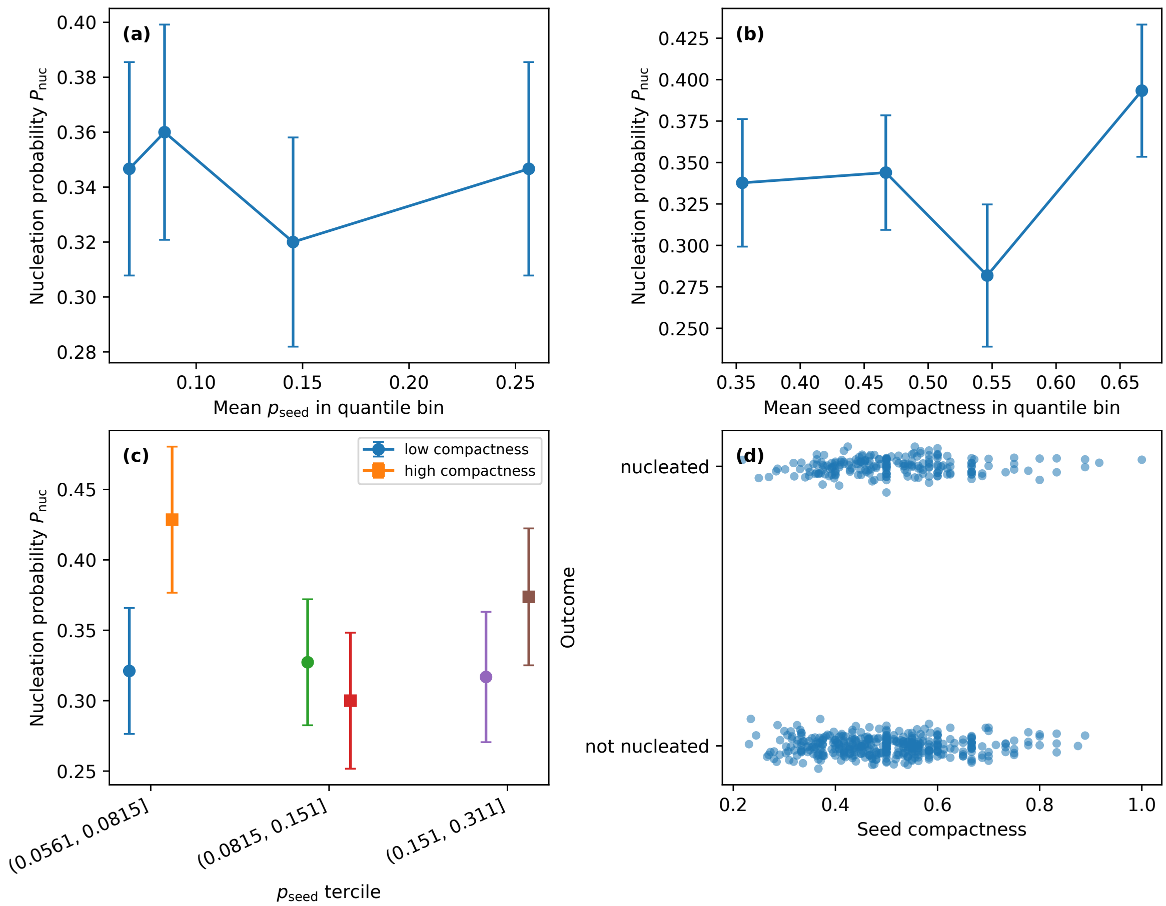

FIG. S5. Sample-level seed-quality diagnostics for the main first-order data set. (a) Nucleation probability binned by p_{seed}. (b) Nucleation probability binned by seed compactness. (c) Low- and high-compactness subsets compared within p_{seed} terciles. (d) Jittered binary nucleation outcome against compactness. These diagnostics provide only modest predictive support for compactness and are used as a conservative consistency check rather than as a committor analysis.

TABLE S3. Bootstrap and permutation trend checks for the main first-order data set after adding the perimeter-to-area diagnostic. The Spearman coefficient is computed from the six label-averaged values. Bootstrap confidence intervals are obtained by resampling realizations within each label.

Observable	Spearman ρ	Permutation p	Bootstrap 95% CI	High–low difference
p_{seed}	1.000	0.0023	$[1.000, 1.000]$	0.219
c_{seed}	1.000	0.0034	$[0.943, 1.000]$	31.62
C_{comp}	-1.000	0.0028	$[-1.000, -0.943]$	-0.184
R_g	1.000	0.0029	$[0.943, 1.000]$	2.214
$P_{\mathrm{seed}}/A_{\mathrm{seed}}$	-1.000	0.0031	$[-1.000, -0.943]$	-0.369

BOOTSTRAP AND PERMUTATION TREND CHECKS FOR SEED GEOMETRY INCLUDING PERIMETER-TO-AREA

We additionally checked the label-level trends in the main seed observables by bootstrap resampling within each initial-state label and by a permutation test of the label-mean trend. The test statistic is the Spearman rank correlation between the initial-state label and the label-averaged observable. For the largest barrier-crossing seed cluster, we also compute a lattice perimeter-to-area diagnostic, $P_{\mathrm{seed}}/A_{\mathrm{seed}}$, where the perimeter is the number of nearest-neighbor boundary edges on the periodic square lattice. This ratio depends on both shape and size: it is sensitive not only to boundary irregularity but also to the growth of cluster area. This analysis is not intended to prove a microscopic nucleation mechanism. Its purpose is to test whether the trends shown in Figs. 3 and 4 are stable under resampling of the finite ensemble.

Table S3 summarizes the results. The trends are very stable in the present data set. The seed amount measures p_{seed} and c_{seed} increase monotonically with the initial-state label, while the compactness decreases monotonically, the radius of gyration increases monotonically, and the perimeter-to-area diagnostic decreases monotonically. In all four cases, the bootstrap confidence interval for the Spearman trend does not cross zero.

Figure S6 shows the corresponding label-mean trends. These results strengthen the conservative conclusion that seed amount and seed geometry change systematically in different directions. The perimeter-to-area diagnostic decreases with the initial-state label, while the absolute perimeter increases. This indicates that larger-label seeds are not simply compact droplets with larger interface length; rather, the largest barrier-crossing clusters become larger and more spatially extended, with a lower perimeter-to-area value because their area grows faster than their boundary length. Therefore, the perimeter-to-area result should be interpreted together with compactness and R_g, not as a standalone committor-like predictor.

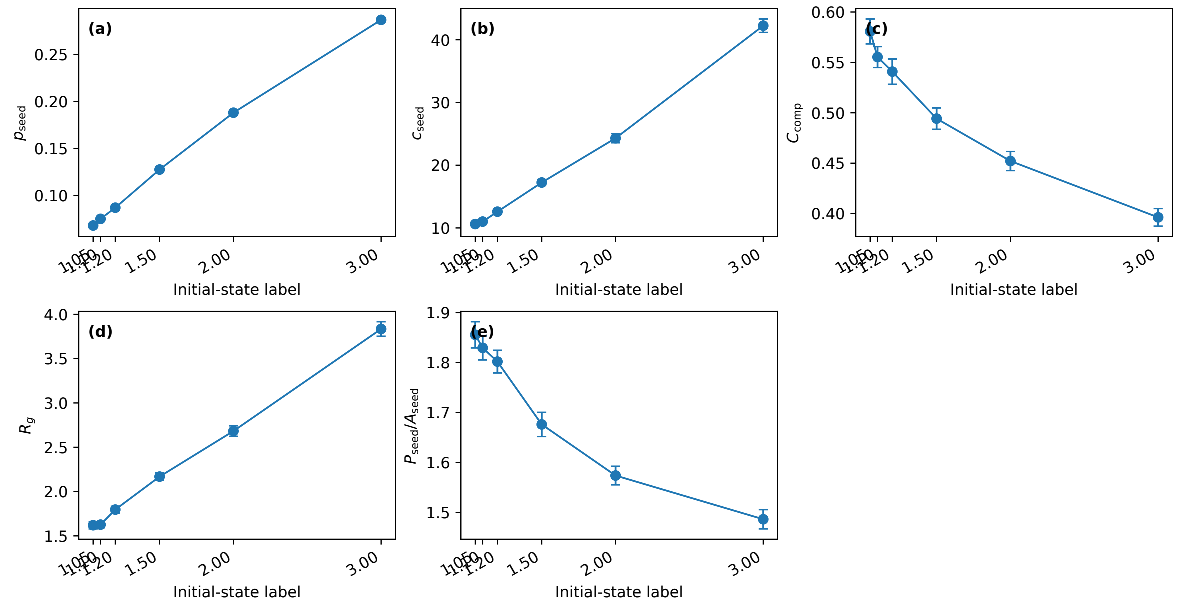

FIG. S6. Bootstrap/permutation trend-check diagnostics for the main first-order data set. The plotted observables are label means with standard-error bars in a compact multi-panel layout. The trends in seed amount, compactness, radius of gyration, and perimeter-to-area are stable under bootstrap resampling within labels.

TABLE S4. Finite-time committor-style restart analysis. For each initial-state label, eight frozen initial configurations were generated and each was restarted sixteen times with independent post-quench noise.

Label	Configurations	Restarts	Mean q_{nuc}	Range of q_{nuc}
1.05	8	128	0.297	0.125–0.438
1.10	8	128	0.336	0.188–0.438
1.20	8	128	0.289	0.125–0.438
1.50	8	128	0.281	0.063–0.563
2.00	8	128	0.344	0.188–0.500
3.00	8	128	0.375	0.188–0.688

FINITE-TIME COMMITTOR-STYLE RESTART ANALYSIS

To test more directly whether the initial seed observables predict nucleation, we performed a finite-time restart analysis. For each selected initial configuration ϕ_0, the post-quench dynamics was restarted multiple times with independent post-quench noise. We then estimated

$$q_{\mathrm{nuc}}(\phi_0) = \frac{\text{number of restarts that nucleate by } t_{\mathrm{max}}}{\text{number of restarts}}, \tag{2}$$

using the same cluster-nucleation criterion as in the main analysis. This quantity is not an exact transition-path committor because the observation time is finite and the event definition is operational. It is nevertheless a direct restart-based diagnostic of whether a given initial configuration is more or less likely to nucleate under the post-quench dynamics.

The finite-time restart analysis used eight frozen initial configurations for each of the six initial-state labels and sixteen independent post-quench restarts per configuration, for a total of 768 restart trajectories. Table S4 summarizes the label-averaged values of q_{nuc}. The values lie between approximately 0.28 and 0.38, with substantial configuration-to-configuration variation within each label.

Table S5 reports correlations between q_{nuc} and several initial seed observables across the 48 frozen initial configurations. The correlations are weak for all tested low-dimensional observables. In particular, the Pearson and Spearman correlations with seed compactness, radius of gyration, perimeter, and perimeter-to-area are small in magnitude. Thus, within

TABLE S5. Correlations between the finite-time committor-like probability q_{nuc} and initial seed observables across the 48 frozen initial configurations.

Observable	Pearson correlation	Spearman correlation
p_{seed}	0.185	0.131
c_{seed}	0.007	-0.029
C_{comp}	-0.144	-0.144
R_g	0.028	-0.010
Seed perimeter	0.036	0.006
Seed perimeter/area	0.042	0.125
p_{ordered}	0.017	-0.064
c_{ordered}	-0.014	-0.066

this finite restart ensemble, the simple seed-geometry observables used in the main text are not sufficient committor-level predictors.

Figure S7 visualizes the same restart analysis. The weak correlations should not be read as a failure of the seed-geometry diagnostics. Rather, they delimit their interpretation. Compactness, radius of gyration, and perimeter-to-area capture systematic label-level changes in the morphology of barrier-crossing seed clusters, but a finite-time nucleation probability depends on additional dynamical information not contained in these few scalar descriptors alone. This supports the conservative framing used in the main text: seed amount and seed geometry should be tracked separately, but the present geometry measures are not a complete committor theory.

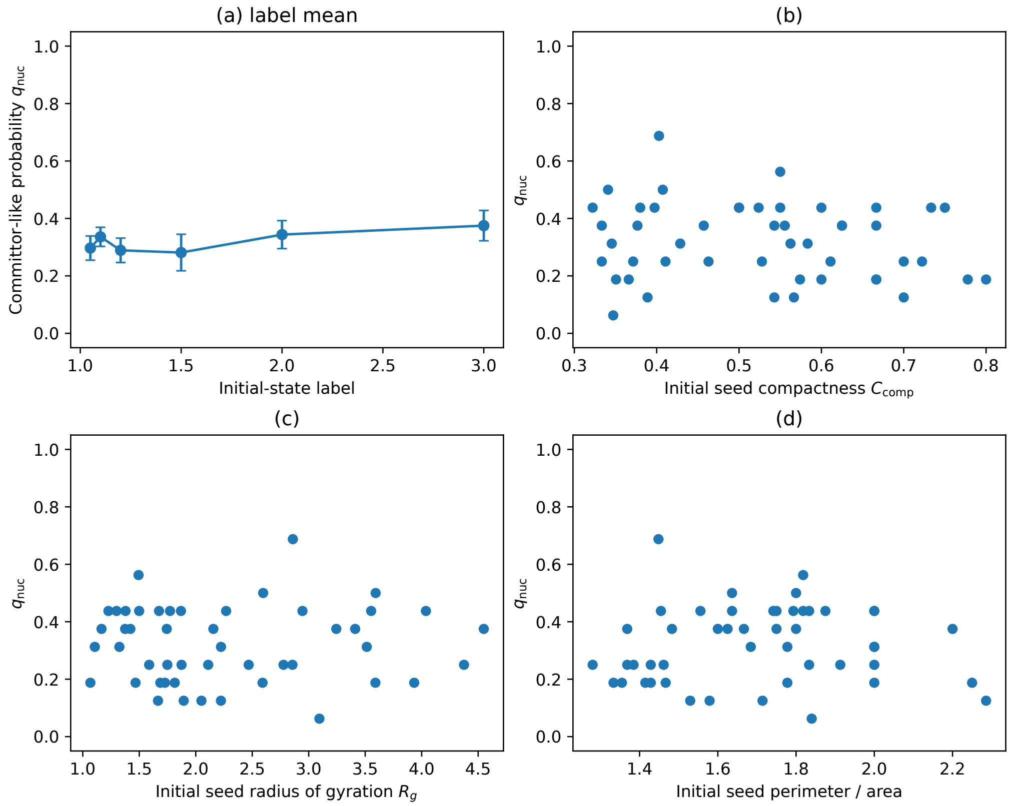

FIG. S7. Finite-time committor-style restart analysis. (a) Label-averaged restart probability q_{nuc}. (b–d) Configuration-level relations between q_{nuc} and initial seed compactness, radius of gyration, and perimeter-to-area. The weak correlations indicate that these simple scalar geometry diagnostics are not sufficient committor-level predictors in the present finite restart ensemble.